

Doubling a boundary problem and computing its index

A first-order elliptic boundary problem can be turned into an operator on a closed doubled manifold. The construction has three parts that must agree exactly: the reflected half must carry the complementary stable modes, the boundary coupling must stay elliptic while separate boundary conditions become matching traces, and the resulting mixed operator must be approximated in operator norm before its continuous symbol can compute the index.

This lesson proves all three parts. It also resolves a possible factor-of-two ambiguity: the geometric double built here has a zero-index complementary half, so its index equals the boundary index. A one-half formula belongs to a different doubled datum whose complementary half has the same index as the original problem.

The named prerequisites are Stable modes and the algebra of boundary data, Fredholm boundary problems with first-order Calderón defects, Finite defects under perturbation, and Symbols, finite defects, and the index on a closed manifold. We use $D_t = -i\partial_t$, inward collar coordinates, and Hermitian inner products linear in the first entry. Matrix factors retain their displayed order.

1. The split first-order model

Let X be compact with boundary Y . Suppose

$$E|_Y = E^+ \oplus E^- \quad (\text{DI1})$$

is orthogonal for a chosen Hermitian metric. Choose a product density and product metric in a collar $Y \times [0, \delta)_t$. Let $\phi \in C_c^\infty([0, \delta))$ be real, $0 \leq \phi \leq 1$, and $\phi = 1$ near $t = 0$. Take $\Lambda^\pm \in \Psi_{\text{phg}}^1(Y; E^\pm)$ with positive scalar real principal symbols $\lambda^\pm(y, \eta)I_{E^\pm}$, and take an interior operator $\Lambda \in \Psi_{\text{phg}}^1(X^\circ; E)$ whose principal symbol $\lambda(x, \xi)I_E$ is positive for every $\xi \neq 0$. Extend $\Lambda^\pm$ constantly in t where $\phi = 1$, and put

$$\begin{aligned} P^b u &= \phi \begin{pmatrix} D_t + i\Lambda^+ & 0 \\ 0 & -D_t + i\Lambda^- \end{pmatrix} \begin{pmatrix} u^+ \\ u^- \end{pmatrix}, \\ P^i u &= i(1 - \phi)\Lambda(1 - \phi)u, \quad P = P^b + P^i, \quad Bu = \gamma_0 u^-. \end{aligned} \quad (\text{DI2})$$

In the collar-transition region the principal symbol is

$$p(y, t, \eta, \tau) = \phi(t) \begin{pmatrix} \tau + i\lambda^+(y, \eta) & 0 \\ 0 & -\tau + i\lambda^-(y, \eta) \end{pmatrix} + i(1 - \phi(t))^2 \lambda(y, t, \eta, \tau) I_E. \quad (\text{DI3})$$

If $0 < \phi < 1$, the imaginary part of each diagonal scalar is positive at every nonzero covector. Where $\phi = 1$, a nonzero tangential covector gives a positive imaginary part and a nonzero pure normal covector gives the real entries $\pm\tau$. Where $\phi = 0$, the last term is invertible. Thus P is elliptic.

Freeze at $(y, \eta) \in T^*Y \setminus 0$. The two normal equations and their solutions are

$$\begin{aligned} (D_t + i\lambda^+)v^+ &= 0, \quad v^+(t) = e^{\lambda^+ t} v^+(0), \quad v^+ \text{ bounded} \iff v^+(0) = 0, \\ (-D_t + i\lambda^-)v^- &= 0, \quad v^-(t) = e^{-\lambda^- t} v^-(0), \quad v^- \text{ bounded for every } v^-(0). \end{aligned} \quad (\text{DI4})$$

Hence the stable Cauchy space is exactly E_y^- , and the principal boundary map $v \mapsto v^-(0)$ is its identity. The generalized interior and boundary symbols are elliptic. Fredholm boundary problems with first-order Calderón defects therefore makes

$$(P, B)_s : \bar{H}_s(X^\circ; E) \longrightarrow \bar{H}_{s-1}(X^\circ; E) \oplus H^{s-1/2}(Y; E^-) \quad (\text{DI5})$$

Fredholm for every real $s \geq 1$.

2. The boundary signs and the shifted kernel

For a compactly supported smooth scalar or vector function w in the collar, integration by parts with the stated inner-product convention gives the exact signs

$$\begin{aligned} 2 \operatorname{Im}(\phi D_t w, w)_X &= \|\gamma_0 w\|_Y^2 + \int_X \phi'(t) |w|^2, \\ 2 \operatorname{Im}(-\phi D_t w, w)_X &= -\|\gamma_0 w\|_Y^2 - \int_X \phi'(t) |w|^2. \end{aligned} \quad (\text{DI6})$$

Indeed, the first difference from its complex conjugate is $-i \int \phi \partial_t |w|^2$, and the endpoint at $t = 0$ has the displayed positive sign. The second identity is its negative.

The scalar positive principal symbols of $\Lambda^\pm$ and Λ give the matrix Gårding lower bounds for the real parts of their quadratic forms. Derivatives of ϕ , nonsymmetric lower-order terms, and the bounded collar patching terms contribute at most $C\|u\|_X^2$. Combining these facts with (DI6) gives

$$2 \operatorname{Im}(Pu, u)_X \geq \|\gamma_0 u^+\|_Y^2 - \|\gamma_0 u^-\|_Y^2 - C\|u\|_X^2. \quad (\text{DI7})$$

For $T > C/2$, set $P_T = P + iTI_E$. If $P_T u = 0$ and $Bu = 0$, then

$$0 = 2 \operatorname{Im}(P_T u, u)_X \geq \|\gamma_0 u^+\|_Y^2 + (2T - C)\|u\|_X^2. \quad (\text{DI8})$$

Thus $u = 0$. Elliptic regularity from Fredholm boundary problems with first-order Calderón defects first makes every weak kernel element smooth, so the calculation applies to the full kernel of (DI5).

3. The adjoint boundary relation and the shifted cokernel

Take the base realization $s = 1$. A pair (v, h) in the annihilator of the range of $(P_T, B)_1$ is smooth by the dual regularity theorem in Fredholm boundary problems with first-order Calderón defects. The leading normal terms give Green's identity

$$(Pu, v)_X - (u, P^*v)_X = i(\gamma_0 u^+, \gamma_0 v^+)_Y - i(\gamma_0 u^-, \gamma_0 v^-)_Y. \quad (\text{DI9})$$

The annihilation equation is therefore

$$0 = (u, (P^* - iT)v)_X + i(\gamma_0 u^+, \gamma_0 v^+)_Y - i(\gamma_0 u^-, \gamma_0 v^-)_Y + (\gamma_0 u^-, h)_Y. \quad (\text{DI10})$$

Interior tests and arbitrary boundary traces give, with no change of the conjugate-linear slot,

$$(P^* - iT)v = 0, \quad \gamma_0 v^+ = 0, \quad h = -i\gamma_0 v^-. \quad (\text{DI11})$$

Put $Q = -P^*$. Exchange the names of the two orthogonal summands:

$$E_Q^+ = E^-, \quad E_Q^- = E^+. \quad (\text{DI12})$$

The full differentiated-cutoff terms are computed below in (DA3)–(DA5). The collar principal part of Q , written in the order $E_Q^+ \oplus E_Q^-$, is

$$\begin{pmatrix} D_t + i(\Lambda^-)^* & 0 \\ 0 & -D_t + i(\Lambda^+)^* \end{pmatrix}, \quad B_Q v = \gamma_0 v^+ = \gamma_0 v_Q^-. \quad (\text{DI13})$$

Multiplying the first equation in (DI11) by -1 gives $(Q + iT)v = 0$, while $B_Q v = 0$. The estimate (DI8), with the same sufficiently large T after enlarging C once, gives $v = 0$, and then $h = 0$. Thus $(P_T, B)_1$ is bijective. The path

$$(P + irTI_E, B), \quad 0 \leq r \leq 1, \quad (\text{DI14})$$

keeps both principal symbols fixed and stays Fredholm. Its index is constant, so $\text{ind}(P, B)_1 = 0$. Fredholm boundary problems with first-order Calderón defects identifies the same smooth kernel and adjoint obstruction spaces at every $s \geq 1$. Consequently

$$\boxed{\text{ind}(P, B)_s = 0 \quad (s \geq 1).} \quad (\text{DI15})$$

The full positivity remainder and adjoint coefficients

Here is a direct proof of the lower bounds used in (DI7), with their actual lower-order operators retained. If A is any of Λ^+ , Λ^- , or the interior Λ , its principal symbol is $\lambda_A I$, with $\lambda_A > 0$ off the zero section. Quantize the degree-one-half symbol $\sqrt{\lambda_A} I$ to an operator R_A . Use the full bundle composition and adjoint formulas in Symbols, operators and Sobolev scales and Detecting regularity without choosing coordinates. Their principal product is exactly $\lambda_A I$. Hence

$$H_A = \frac{A + A^*}{2} - R_A^* R_A \in \Psi^0, \quad \text{Re}(Aw, w) = \|R_A w\|_2^2 + (H_A w, w), \quad C_A = \|H_A\|_{L^2 \rightarrow L^2}. \quad (\text{DA1})$$

The principal symbol comparison proves the order-zero claim; the linked order-zero mapping theorem makes C_A finite. The formula retains the entire lower-order operator H_A , rather than replacing A by its positive principal symbol. It gives $\text{Re}(Aw, w) \geq -C_A \|w\|_2^2$. For tangential families take the supremum of these constants on the compact support of ϕ . Smooth family quantization and the finite seminorm bounds give finite suprema. For the interior operator the input $(1 - \phi)u$ is supported away from the boundary; insert cutoffs equal to one on that support and use the same calculation on interior charts. All errors from these cutoffs are included in H_Λ .

At each fixed t , the scalar $\phi(t)$ commutes with the tangential operators. It is nonnegative, so multiplying (DA1) by it and integrating in t gives the tangential lower bound without taking a square root of the cutoff. The exact interior quadratic form is $\text{Im}(i(1 - \phi)\Lambda(1 - \phi)u, u) = \text{Re}(\Lambda(1 - \phi)u, (1 - \phi)u)$. Combining these statements with both full identities in (DI6) proves (DI7) with the concrete choice

$$C_b = \max(C_{\Lambda^+}, C_{\Lambda^-}), \quad C_i = C_\Lambda, \quad C_P = \|\phi'\|_\infty + 2C_b + 2C_i. \quad (\text{DA2})$$

No self-adjointness of A was assumed. Its geometric adjoint has the same real quadratic form, so the same constants work for Λ^* and $(\Lambda^\pm)^*$.

To compute $Q = -P^*$, use the original product metric and density. Multiplication by the real ϕ is self-adjoint and $D_t\phi = \phi D_t - i\phi'$. The full collar adjoint, before exchanging summands, is

$$(P^b)^* = \begin{pmatrix} \phi D_t - i\phi' - i\phi(\Lambda^+)^* & 0 \\ 0 & -\phi D_t + i\phi' - i\phi(\Lambda^-)^* \end{pmatrix}, \quad (P^i)^* = -i(1 - \phi)\Lambda^*(1 - \phi). \quad (\text{DA3})$$

Tangential operators commute with $\phi(t)$, including when their coefficients depend on t . Let $S(v^+, v^-) = (v^-, v^+)$ be the unitary change of order on the collar. In precisely the order (DI12) the full operator is

$$\begin{aligned} SQ^b S^{-1} &= \phi \begin{pmatrix} D_t + i(\Lambda^-)^* & 0 \\ 0 & -D_t + i(\Lambda^+)^* \end{pmatrix} + \begin{pmatrix} -i\phi' & 0 \\ 0 & i\phi' \end{pmatrix}, \\ SQ^i S^{-1} &= i(1 - \phi)S\Lambda^* S^{-1}(1 - \phi), \\ B_Q v &= \gamma_0 v^+ = \gamma_0 (Sv)^-. \end{aligned} \quad (\text{DA4})$$

The second formula is a collar expression only; it does not assert that the splitting or S extends over the whole interior bundle. On the original bundle the full interior operator is $i(1 - \phi)\Lambda^*(1 - \phi)$. These descriptions give the same quadratic form on their respective charts. The displayed multiplication remainder contributes exactly

$$2 \operatorname{Im} \left(\begin{pmatrix} -i\phi' & 0 \\ 0 & i\phi' \end{pmatrix} v_Q, v_Q \right) = -2 \int \phi' |v_Q^+|^2 + 2 \int \phi' |v_Q^-|^2. \quad (\text{DA5})$$

Its absolute value is at most $2\|\phi'\|_\infty \|v\|_2^2$. Therefore (DI7) holds for Q with $C_Q = C_P + 2\|\phi'\|_\infty$, with its plus and minus boundary spaces exactly as in (DI12). A single $T > \max(C_P, C_Q)/2$ supplies both estimates used in (DI8) and (DI11). This proves the shifted bijectivity without discarding either differentiated-cutoff term.

4. A surjective boundary constraint preserves the index

Let X_0, Y_0, Z_0 be Hilbert spaces, let $A = (P, B) : X_0 \rightarrow Y_0 \oplus Z_0$ be Fredholm, and suppose $B : X_0 \rightarrow Z_0$ is surjective. Its restriction to $(\ker B)^\perp$ is a bounded bijection onto Z_0 , so the Banach inverse theorem supplies a bounded right inverse $R : Z_0 \rightarrow X_0$. Set $X_B = \ker B$. Define

$$\begin{aligned} J : X_B \oplus Z_0 &\longrightarrow X_0, & J(x, z) &= x + Rz, \\ J^{-1}u &= (u - RBu, Bu), \\ S : Y_0 \oplus Z_0 &\longrightarrow Y_0 \oplus Z_0, & S(y, z) &= (y - PRz, z). \end{aligned} \quad (\text{DI16})$$

Both J and S are bounded isomorphisms, and direct substitution gives the exact triangular reduction

$$SAJ(x, z) = (Px, z), \quad SAJ = (P|_{X_B}) \oplus I_{Z_0}. \quad (\text{DI17})$$

It follows that $P|_{X_B} : X_B \rightarrow Y_0$ is Fredholm and that its kernel, cokernel, and index agree with those of A :

$$\ker(P|_{X_B}) = \ker A, \quad \operatorname{coker}(P|_{X_B}) \simeq \operatorname{coker} A, \quad \operatorname{ind}(P|_{X_B}) = \operatorname{ind} A. \quad (\text{DI18})$$

This also proves the cokernel isomorphism, rather than only an index count.

5. The reflected complementary half

Now let $P_1 : E_1 \rightarrow F_1$ be any elliptic first-order mixed operator on a copy $X_1 = X$ whose collar part, after a fixed collar identification $\kappa : F_1 \simeq E_1$, is the split expression in (DI2). Its compactly supported interior part need not be the positive model used to prove (DI15). Give a second copy $X_2 = X$ the inward coordinate $t_2 \geq 0$, copy the split bundle $E_2^+ \oplus E_2^-$, and set near its boundary

$$P_2^b = \phi \begin{pmatrix} -D_{t_2} + i\Lambda^+ & 0 \\ 0 & D_{t_2} + i\Lambda^- \end{pmatrix}, \quad B_2 u_2 = \gamma_0 u_2^+. \quad (\text{DI19})$$

Complete P_2^b away from the boundary by the reflected version of the positive interior term in (DI2): choose $\Lambda_2 \in \Psi_{\text{phg}}^1(X_2^c; E_2)$ with positive scalar principal symbol and set $P_2^i = i(1 - \phi)\Lambda_2(1 - \phi)$, $P_2 = P_2^b + P_2^i$. Its frozen solutions are

$$u_2^+(t_2) = e^{-\lambda^+ t_2} u_2^+(0), \quad u_2^-(t_2) = e^{\lambda^- t_2} u_2^-(0). \quad (\text{DI20})$$

Thus its stable trace is exactly E^+ , and B_2 is the identity there. After ordering the summands as $E_2^- \oplus E_2^+$, this is the model of Section 1. Hence

$$\operatorname{ind}(P_2, B_2) = 0. \quad (\text{DI21})$$

6. Smooth gluing of the doubled bundles and operator

Glue X_1 and X_2 along Y . In the resulting two-sided collar use the signed coordinate

$$r = t_1 \quad \text{on } X_1, \quad r = -t_2 \quad \text{on } X_2. \quad (\text{DI22})$$

Glue E_1 to E_2 by the identity on $E^+ \oplus E^-$, producing $\widehat{E}$. Glue F_1 to the target E_2 by the collar identification κ , producing $\widehat{F}$. Since $D_{t_2} = -D_r$, both collar formulas become

$$\widehat{P}^b = \begin{pmatrix} D_r + i\Lambda^+ & 0 \\ 0 & -D_r + i\Lambda^- \end{pmatrix} \quad (\text{DI23})$$

on a full two-sided neighborhood of the seam. The interior terms vanish on that neighborhood because $1 - \phi = 0$ there. Thus all coefficient jets and the bundle maps match, and the piecewise operator defines

$$\widehat{P} : H^s(\widehat{X}; \widehat{E}) \longrightarrow H^{s-1}(\widehat{X}; \widehat{F}). \quad (\text{DI24})$$

This is a smooth mixed collar operator. It need not be an ordinary global pseudodifferential operator, because $\Lambda^\pm$ act only in the tangential variables near the seam. This proves both the construction and the distinction between a mixed collar operator and an ordinary global pseudodifferential operator.

7. The coupling path and the H^1 gluing domain

At the base level put

$$\begin{aligned}\mathcal{X} &= \bar{H}_1(X_1; E_1) \oplus \bar{H}_1(X_2; E_2), \\ \mathcal{Y} &= L^2(X_1; F_1) \oplus L^2(X_2; E_2), \\ \mathcal{Z} &= H^{1/2}(Y; E^+) \oplus H^{1/2}(Y; E^-), \\ C_\tau(u_1, u_2) &= (\gamma_0 u_2^+ - \tau \gamma_0 u_1^+, \gamma_0 u_1^- - \tau \gamma_0 u_2^-), \quad 0 \leq \tau \leq 1, \\ \mathcal{A}_\tau &= (P_1 \oplus P_2, C_\tau) : \mathcal{X} \longrightarrow \mathcal{Y} \oplus \mathcal{Z}.\end{aligned}\tag{DI25}$$

The stable traces of $P_1 \oplus P_2$ have the form $(0, a^-; a^+, 0)$. On them,

$$C_\tau(0, a^-; a^+, 0) = (a^+, a^-),\tag{DI26}$$

independently of τ . The principal boundary map is therefore an isomorphism for every parameter. The generalized Fredholm theorem and homotopy stability give

$$\text{ind } \mathcal{A}_\tau = \text{ind } \mathcal{A}_0 \quad (0 \leq \tau \leq 1).\tag{DI27}$$

The coupling map itself is surjective. For data (a^+, b^-) , choose fixed Sobolev trace extensions with

$$\gamma_0 u_2^+ = a^+, \quad \gamma_0 u_1^- = b^-, \quad \gamma_0 u_1^+ = 0, \quad \gamma_0 u_2^- = 0.\tag{DI28}$$

This gives one bounded right inverse $R : \mathcal{Z} \rightarrow \mathcal{X}$ for every C_τ . Apply (DI18):

$$\text{ind}((P_1 \oplus P_2)|_{\ker C_\tau}) = \text{ind } \mathcal{A}_\tau.\tag{DI29}$$

At $\tau = 0$, the two constraints are $\gamma_0 u_2^+ = 0$ and $\gamma_0 u_1^- = 0$. After a fixed permutation of the target factors, the constrained operator is the direct sum of $P_1|_{\ker B_1}$ and $P_2|_{\ker B_2}$. Equation (DI18) identifies their indices with those of (P_1, B_1) and (P_2, B_2) . Equations (DI21), (DI27), and (DI29) give

$$\text{ind}((P_1 \oplus P_2)|_{\ker C_1}) = \text{ind}(P_1, B_1).\tag{DI30}$$

The varying kernels can also be trivialized explicitly. Put

$$K(u_1, u_2) = (\gamma_0 u_1^+, \gamma_0 u_2^-), \quad C_\tau = C_0 - \tau K, \quad KR = 0, \quad C_0 R = I_{\mathcal{Z}}.\tag{DI31}$$

For $x \in \ker C_0$, define

$$J_\tau x = x + \tau R K x.\tag{DI32}$$

Then $C_\tau J_\tau x = 0$. Its inverse on $\ker C_\tau$ is $y \mapsto y - R C_0 y$, because $C_0 y = \tau K y$ and $KR = 0$. Hence $J_\tau : \ker C_0 \rightarrow \ker C_\tau$ is a continuous family of bounded isomorphisms. This makes the Fredholm deformation in (DI30) an actual fixed-domain path, not merely an index comparison between unnamed spaces.

At $\tau = 1$, the constraint is equality of the two full boundary traces. The exact H^1 gluing identity is

$$H^1(\widehat{X}; \widehat{E}) \simeq \{(u_1, u_2) \in \mathcal{X} : \gamma_0 u_1^+ = \gamma_0 u_2^+, \gamma_0 u_1^- = \gamma_0 u_2^-\} = \ker C_1. \quad (\text{DI33})$$

One direction follows by restriction. Conversely, the distributional first derivative of a piecewise H^1 section has a seam delta whose coefficient is its trace jump; equality of traces removes that delta, leaving every weak first derivative in L^2 . Under (DI33), the constrained piecewise operator is exactly (DI24). Thus

$$\boxed{\text{ind } \widehat{P} = \text{ind}(P_1, B_1)} \quad (\text{DI34})$$

at $H^1 \rightarrow L^2$. The statement at this step is deliberately made at $s = 1$: equality of value traces alone is not the complete gluing condition for arbitrary higher Sobolev order.

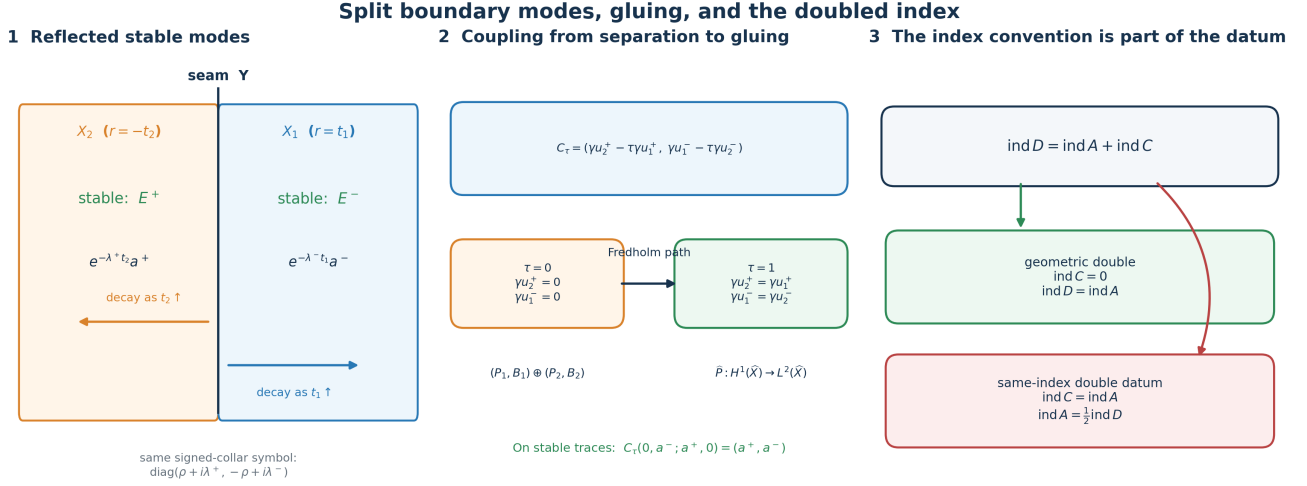


Figure 1: The stable summands on the reflected halves, the coupling path, and the two index strata

The left panel keeps both inward coordinates and the signed collar coordinate visible. The middle panel shows why the boundary symbol stays invertible throughout the coupling. The right panel records the complementary index that distinguishes equality from a one-half formula.

The complete seam distribution

We verify the asserted gluing in the actual product collar. Use its signed coordinate $r = t_1 = -t_2$, product density $d\mu_Y dr$, and the domain-bundle identification from Section 6. In any product frame, write

$$u(y, r) = \begin{cases} u_1(y, r), & r > 0, \\ u_2(y, -r), & r < 0, \end{cases} \quad j(y) = \gamma_0 u_1(y) - \gamma_0 u_2(y). \quad (\text{DG1})$$

Let H be the Heaviside function; its value at zero does not affect the distribution. Integration by parts on each half, with the common boundary measure, gives every first derivative:

$$\begin{aligned} D_r u &= H(r)D_{t_1} u_1(y, r) - H(-r)D_{t_2} u_2(y, -r) - ij(y)\delta(r), \\ D_{y_a} u &= H(r)D_{y_a} u_1(y, r) + H(-r)D_{y_a} u_2(y, -r), \quad 1 \leq a \leq \dim Y. \end{aligned} \tag{DG2}$$

For smooth half-sections this follows directly from $\partial_r H(r) = \delta(r)$ and $\partial_r H(-r) = -\delta(r)$. For half-sections in H^1 , approximate on each half by smooth sections. The continuous $H^1 \rightarrow H^{1/2}$ trace map from Mixed symbols on every real two-parameter Sobolev scale makes the seam coefficients converge, while the half-space functions and weak derivatives converge in L^2 . Pairing with each compactly supported smooth test section passes the identities to the limit. Thus no differentiability of the traces is assumed.

All bulk terms in (DG2) are in L^2 . If $j = 0$, every weak first derivative is in L^2 , proving the converse in (DI33). If $j \neq 0$, choose a smooth compactly supported tangential test section b with $\langle j, b \rangle \neq 0$. Such a test exists because a nonzero distribution cannot vanish on every test section. Choose $h \in C_c^\infty(\mathbb{R})$ with $h(0) = 1$, and put $h_\varepsilon(r) = h(r/\varepsilon)$. Then

$$\begin{aligned} \langle j\delta(r), b(y)h_\varepsilon(r) \rangle &= \langle j, b \rangle, \\ \|b(y)h_\varepsilon(r)\|_{L^2(d\mu_Y dr)} &= \varepsilon^{1/2} \|b\|_{L^2(d\mu_Y)} \|h\|_{L^2(dr)}. \end{aligned} \tag{DG3}$$

An L^2 distribution has pairings bounded by its L^2 norm times the second line, which tends to zero. The first line is a fixed nonzero number. Therefore the seam delta is not in L^2 . In (DG2) it cannot cancel an L^2 bulk term. This proves necessity of trace equality as well.

The restriction map and the piecewise inverse are bounded for the finite-chart H^1 norms: (DG2) with zero jump bounds every coordinate first derivative by the two half-norms, and restriction bounds each half-norm by the global norm. Smooth transition matrices contribute their actual bounded first derivatives; the fixed product metric and density compare these finitely many chart norms. This proves the bounded isomorphism (DI33), including its full domain. It makes no claim that matching values alone glues higher Sobolev orders, which would require the corresponding normal-jet conditions.

8. Approximation by ordinary operators and the symbol index

In the signed collar let (η, ρ) be the covariables dual to (y, r) . The principal symbol of (DI23), extended by the ordinary interior symbols, is continuous and degree one on $T^*\widehat{X} \setminus 0$. At $\eta = 0$, positive homogeneity gives $\lambda^\pm(y, 0) = 0$, so its two collar blocks equal ρ and $-\rho$. Thus it is still invertible there. Denote this continuous elliptic symbol by

$$\widehat{p} : T^*\widehat{X} \setminus 0 \longrightarrow \text{Hom}(\pi^*\widehat{E}, \pi^*\widehat{F}). \tag{DI35}$$

We now verify the operator-norm approximation required by Symbols, finite defects, and the index on a closed manifold rather than inferring it from pointwise smoothing. Choose the scalar cutoff from Symbols, finite defects, and the index on a closed manifold, Section 10,

$$\chi(\eta, \rho) = 1 \text{ if } |\rho| \leq \max(1, |\eta|), \quad \chi(\eta, \rho) = 0 \text{ if } |\rho| \geq 2\max(1, |\eta|), \tag{DI36}$$

and for $0 < \varepsilon \leq 1$ replace every partial tangential symbol $a(y, r, \eta)$ of order one by

$$a_\varepsilon(y, r, \eta, \rho) = a(y, r, \eta)\chi(\eta, \varepsilon\rho). \tag{DI37}$$

For fixed $\varepsilon > 0$, this is a classical full-variable symbol. On the total unit cosphere its leading term differs from the continuous zero extension of the partial leading symbol only where $|\eta| \leq 2\varepsilon|\rho|$. Since that leading term has size at most $C|\eta|$,

$$\sup_{|\eta|^2 + \rho^2 = 1} \|a_{\varepsilon,1} - a_1\| \leq C\varepsilon. \quad (\text{DI38})$$

The exact partial-operator estimate in Section 10 of Symbols, finite defects, and the index on a closed manifold gives, for every real s ,

$$\|\text{Op}(a_\varepsilon) - a(y, r, D_y)\|_{H^s \rightarrow H^{s-1}} \leq C_s \varepsilon. \quad (\text{DI39})$$

Apply this construction in a finite signed-collar atlas. Keep D_r and every ordinary interior operator unchanged, and patch with the same fixed bundle charts and cutoffs. The compactness of $\widehat{X}$ turns the local estimates into ordinary classical operators $\widehat{P}_\varepsilon$ satisfying

$$\widehat{P}_\varepsilon \longrightarrow \widehat{P} \quad \text{in } \mathcal{L}(H^s, H^{s-1}) \text{ for every real } s, \quad \widehat{p}_\varepsilon \longrightarrow \widehat{p} \text{ uniformly on } S^* \widehat{X}. \quad (\text{DI40})$$

Uniform invertibility of $\widehat{p}$ on the compact cosphere makes $\widehat{p}_\varepsilon$ elliptic for all sufficiently small ε . The norm-limit theorem and its continuous-symbol identity, proved in Sections 5 and 8 of that lesson, now give

$$\text{ind } \widehat{P} = \text{ind}(\widehat{p}). \quad (\text{DI41})$$

Because (DI40) is compatible at every Sobolev order, the compatible norm-limit conclusion in that lesson identifies the same smooth kernel and adjoint obstruction spaces at all levels. Hence (DI34) and (DI41) hold for every $s \geq 1$:

$$\boxed{\text{ind}(P_1, B_1)_s = \text{ind}(\widehat{P} : H^s \rightarrow H^{s-1}) = \text{ind}(\widehat{p})}. \quad (\text{DI42})$$

This proves the doubled symbol-index formula with the precise approximation hypotheses.

9. The two exact kinds of doubled index datum

Two published index wordings become exact once the complementary half is made part of the datum. Let $A : X_A \rightarrow Y_A$ be Fredholm. A *deformation double datum* for A is a Fredholm operator $D : X_D \rightarrow Y_D$, a Fredholm complement $C : X_C \rightarrow Y_C$, fixed domain and target isomorphisms at the two endpoints, and a continuous fixed-space Fredholm path joining D to $A \oplus C$. Homotopy invariance and direct-sum additivity give the exact morphism from this space of data to the integers:

$$\mathfrak{I}(D, C) = \text{ind } D = \text{ind } A + \text{ind } C. \quad (\text{DI43})$$

There are two mathematically different strata:

$$\begin{aligned} \mathfrak{D}_0(A) &= \{(D, C) : \text{ind } C = 0\} &\implies \text{ind } D &= \text{ind } A, \\ \mathfrak{D}_-(A) &= \{(D, C) : \text{ind } C = \text{ind } A\} &\implies \text{ind } D &= 2 \text{ind } A. \end{aligned} \quad (\text{DI44})$$

Equations (DI31)–(DI34) supply all the data for the geometric double: the fixed-domain path is $(P_1 \oplus P_2)J_\tau$, its endpoint complement is (P_2, B_2) , and (DI21) gives its index zero. Therefore the concrete operator $\widehat{P}$ belongs to $\mathfrak{D}_0(P_1, B_1)$, and its formula is the equality in (DI42).

A one-half formula is valid only after a separately specified double datum is proved to lie in $\mathfrak{D}_=(A)$:

$$\text{ind } A = \frac{1}{2} \text{ind } D \iff \text{ind } C = \text{ind } A \quad \text{within a deformation double datum.} \quad (\text{DI45})$$

Thus the introductory one-half wording cannot be applied to the operator $\widehat{P}$ constructed here. The zero-index complementary half proved in (DI21) gives equality. A summary using one half must name and prove a different same-index complement and its deformation data. Without that information the one-half statement is underdetermined; it is not a replacement for (DI42). This is the exact conceptual correction. It keeps the equality theorem and the conditional one-half formula attached to their respective doubled constructions.

10. Scalar sign check

For $E^+ = E^- = \mathbb{C}$ and $\Lambda^+ = \Lambda^- = |D_y|$, the first-half stable mode is $(0, e^{-|\eta|t_1} a^-)$, while the second-half stable mode is $(e^{-|\eta|t_2} a^+, 0)$. The coupling gives

$$C_\tau(0, e^{-|\eta|t_1} a^-; e^{-|\eta|t_2} a^+, 0) = (a^+, a^-). \quad (\text{DI46})$$

In the signed coordinate, both halves have symbol $\text{diag}(\rho + i|\eta|, -\rho + i|\eta|)$. This checks the reflection sign, the stable summands, the order of the coupling components, and the seam compatibility in one calculation.

11. Three concrete models

A finite-dimensional boundary constraint

Take $X_0 = \mathbb{C}^3$, $Y_0 = \mathbb{C}^2$, $Z_0 = \mathbb{C}$, and

$$P(x_1, x_2, x_3) = (x_1, x_2 + x_3), \quad B(x_1, x_2, x_3) = x_3. \quad (\text{DI47})$$

The combined map (P, B) is invertible. On $\ker B$, the restricted map is $(x_1, x_2, 0) \mapsto (x_1, x_2)$, also invertible. The right inverse $Rz = (0, 0, z)$ makes (DI16)–(DI17) an ordinary triangular matrix factorization. This finite model displays the exact cokernel isomorphism used for Sobolev boundary spaces.

Why matching values is the right H^1 condition

Let $u_1 \in H^1([0, 1])$ and $u_2 \in H^1([-1, 0])$, and join them as a piecewise function u . For every compactly supported smooth test function φ , integration by parts on the two halves gives

$$\langle \partial_r u, \varphi \rangle = \int_{-1}^0 u_2' \varphi \, dr + \int_0^1 u_1' \varphi \, dr + (u_1(0) - u_2(0))\varphi(0). \quad (\text{DI48})$$

The distributional derivative lies in L^2 exactly when the trace jump vanishes. This is the one-dimensional form of (DI33). Higher Sobolev gluing needs further matching derivatives, which is why the index identification is first made at $H^1 \rightarrow L^2$.

The angular cap estimate

For the scalar partial symbol $a(\eta) = |\eta|$, the leading-symbol difference in (DI37) is supported where $|\eta| \leq 2\varepsilon|\rho|$. On $|\eta|^2 + \rho^2 = 1$,

$$|a_{\varepsilon,1} - a_1| \leq |\eta| \leq 2\varepsilon. \quad (\text{DI49})$$

This is the exact small factor that turns the tangential operator into an operator-norm limit of ordinary full-variable pseudodifferential operators.

12. Exercises with complete solutions

1. Recover the two endpoint signs. Starting from $D_t = -i\partial_t$, prove (DI6) without assuming that w is scalar.

Solution. The fiber metric gives $\partial_t|w|^2 = (\partial_t w, w) + (w, \partial_t w)$. Hence

$$\begin{aligned} 2i \operatorname{Im}(\phi D_t w, w) &= -i \int_0^\infty \phi \partial_t |w|^2 dt \\ &= i|w(0)|^2 + i \int_0^\infty \phi' |w|^2 dt. \end{aligned} \quad (\text{DI50})$$

Divide by i . Replacing D_t by $-D_t$ reverses both terms. The calculation is component-free, so it holds for bundle-valued w .

2. Check the cokernel multiplier. In (DI10), use arbitrary minus traces to recover the exact formula for h .

Solution. For $a = \gamma_0 u^-$, the minus terms are $-i(a, \gamma_0 v^-) + (a, h)$. Since the inner product is linear in the first entry, $(a, i\gamma_0 v^-) = -i(a, \gamma_0 v^-)$. Vanishing for every a gives

$$h + i\gamma_0 v^- = 0, \quad h = -i\gamma_0 v^-. \quad (\text{DI51})$$

3. Verify the triangular reduction. Prove directly that the maps J and S in (DI16) are invertible and calculate SAJ .

Solution. Because $BR = I$, $u - RBu \in \ker B$, so the displayed formula for J^{-1} is defined. Direct substitution gives $J^{-1}J(x, z) = (x, z)$ and $JJ^{-1}u = u$. The inverse of S is $S^{-1}(y, z) = (y + PRz, z)$. Finally,

$$SAJ(x, z) = S(Px + PRz, z) = (Px, z). \quad (\text{DI52})$$

4. Trivialize the coupling kernels. Prove that J_τ in (DI32) maps $\ker C_0$ bijectively onto $\ker C_\tau$.

Solution. For $x \in \ker C_0$, use $C_\tau = C_0 - \tau K$, $C_0 R = I$, and $KR = 0$:

$$C_\tau(x + \tau RKx) = 0 + \tau Kx - \tau Kx - \tau^2 K RKx = 0. \quad (\text{DI53})$$

If $y \in \ker C_\tau$, then $C_0 y = \tau Ky$. Put $x = y - RC_0 y$. This lies in $\ker C_0$, $Kx = Ky$, and $x + \tau RKx = y$. Thus $y \mapsto y - RC_0 y$ is the inverse.

5. Test the two index strata. Let $\text{ind } A = -3$. Compute the doubled index for a zero-index complement and for a same-index complement.

Solution. Equation (DI43) gives

$$\text{ind } D = \begin{cases} -3, & \text{ind } C = 0, \\ -6, & \text{ind } C = -3. \end{cases} \quad (\text{DI54})$$

The first construction gives equality. The second gives $\text{ind } A = \frac{1}{2} \text{ind } D = -3$. Moving the factor $1/2$ between these two data would give the wrong integer.

6. Why must the approximation be in operator norm? Explain why pointwise convergence of the full symbols in (DI37) would not by itself prove the index identity.

Solution. Fredholm index stability applies to bounded operators in their operator-norm topology. Pointwise symbol convergence gives no uniform control on the maps between Sobolev spaces and does not exclude a loss concentrated in a shrinking angular cap. The partial-operator estimate supplies the needed bound:

$$\|\widehat{P}_\varepsilon - \widehat{P}\|_{H^s \rightarrow H^{s-1}} \leq C_s \varepsilon. \quad (\text{DI55})$$

Together with uniform principal-symbol convergence and ellipticity, this is exactly the hypothesis of the norm-limit index theorem.

13. Reading notes and references

Equations (DI43)–(DI45) identify the missing datum and give the exact correction for the actual chosen complementary half.

Stable modes and the algebra of boundary data proves the stable-space algebra used in Sections 1 and 5. Fredholm boundary problems with first-order Calderón defects supplies the generalized Fredholm and regularity theorem. Symbols, finite defects, and the index on a closed manifold proves the partial-operator approximation and continuous symbol-index theorem used in Section 8.

The arguments, examples, figure, and exercises in this lesson are independently written. No historical source correspondence beyond the cited statement locations is asserted.

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